

Physics-Guided Multi-Fidelity Residual Learning for Multi-Objective Design of a DC-DC Buck Converter

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Abstract—This paper presents a physics-guided multi-fidelity surrogate workflow for simulation-level multi-objective design of a nominal fixed-duty 12 V-to-5 V buck converter. A low-cost analytical model generated 70,000 Latin-hypercube design points, and the first 300 matched design coordinates were also evaluated with an automated switching-level LTspice reference model; numerical sanity filtering retained 285 simulations. Random Forest, XGBoost, and multilayer-perceptron surrogates fitted the analytical labels well but transferred poorly to LTspice, exposing a cross-fidelity domain shift. Target-wise Gradient Boosting residual models learned the LTspice discrepancy from the retained runs; stored five-fold results were $R^2 \approx 0.724$ for output ripple and $R^2 \approx 0.964$ for total loss, with efficiency MAE ≈ 0.38 percentage points. NSGA-II was stress-tested for extrapolation and then rerun inside the empirical per-variable LTspice envelope with a 1 μ V non-negative ripple safeguard. The final search produced 100 surrogate non-dominated candidates. Three representative candidates were re-evaluated with the original 4–5 ms LTspice protocol, giving efficiency differences of 0.38–0.84 percentage points and ripple differences of 0.92–2.43 mV. A manual reconstruction of the maximum surrogate-predicted-efficiency candidate reproduced the original 4–5 ms result (96.14% efficiency, 58.78 mV ripple), while an alternate 9–10 ms window gave 95.91% efficiency with nearly unchanged ripple, revealing averaging-window sensitivity. The contribution is a transparent cross-fidelity design methodology and optimizer-failure analysis; hardware validation remains future work.

Index Terms—buck converter; multi-fidelity learning; residual correction; surrogate modelling; LTspice; NSGA-II.

I. INTRODUCTION

Machine-learning surrogates can accelerate power-converter design by replacing repeated analytical or circuit-model evaluations during optimization. The main reliability problem is fidelity transfer: a surrogate may fit a simplified analytical model accurately while disagreeing with a switching-level simulation. Evolutionary optimization can amplify that error because it deliberately searches the edges of the learned response surface. This study treats cross-fidelity error and optimizer-driven extrapolation as first-class design problems, rather than only reporting in-domain model accuracy.

The term physics-guided is used deliberately. Converter equations generate the low-fidelity training data and three derived electrical features, but governing equations are not embedded in a PINN-style training loss. The study is also simulation-level: LTspice is used as a higher-detail switching reference relative to the analytical model, not as physical hardware ground truth.

II. RELATED WORK

Prior work already establishes physics-informed converter parameter estimation [1], simulation-trained neural surrogates [2], [3], machine-learning-based component models [4]–[7], automated topology/parameter selection [8], probabilistic feasibility-aware optimization [10], and multi-objective synchronous-buck optimization [11]. Table I summarizes the closest methodological neighbors and the distinction of the present work. Accordingly, no claim is made that machine learning, physics-informed modelling, or converter optimization is itself new.

TABLE I. RELATED-WORK POSITIONING AND DISTINCTION OF THE PRESENT WORK

Study / main approach	Optimization / validation	Distinction of present work
Physics-informed converter modelling [1], [2]	Parameter estimation / time-domain simulation	Present work targets design optimization with analytical→LTspice correction; no PINN loss.
Simulation/component ML surrogates [3]–[7]	Efficiency/component modelling and converter optimization	Explicit matched low-/high-fidelity residual correction at system level.
Wang et al. [8]	Topology + parameter selection	Fixed topology; focus on fidelity transfer and optimizer reliability.
Mahajan et al. [10]	Uncertainty/feasibility-guided optimization	Different probabilistic framework; no analytical low-fidelity residual bridge.
Hernandez et al. [11]	5,992 PLECS cases; multi-objective synchronous-buck search + rechecks	Closest analogue; present work adds explicit cross-fidelity residual correction + extrapolation repair.
Present work	70,000 analytical + 285 matched LTspice; NSGA-II stress test + safeguards + three checks	Audited correction, optimizer-failure diagnosis, and source recoverability.

The resulting novelty is methodological rather than algorithmic. Abundant analytical labels provide low-cost coverage, and matched switching-level labels quantify the fidelity gap; residual models then learn target-wise corrections. NSGA-II serves both as a design engine and as an extrapolation stress test, and the final search combines an empirical LTspice envelope with an explicit non-negative ripple safeguard before representative simulation re-evaluation.

III. METHODOLOGY

Verified physics-guided multi-fidelity workflow

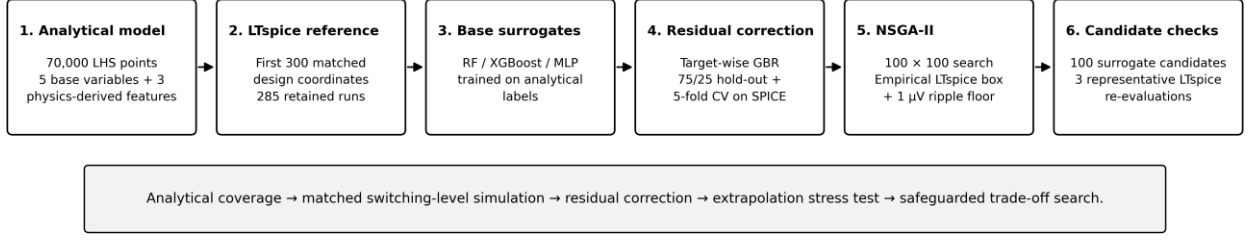


Fig. 1. Source-audited end-to-end workflow used in the publication analysis.

A. Analytical Buck-Converter Model

The nominal design target is $V_{in}=12$ V and $V_{out,target}=5$ V, giving an ideal open-loop duty ratio $D=5/12$. Here, 5 V is the nominal analytical target; the switching-level study uses a fixed open-loop duty ratio and does not optimize closed-loop output-voltage regulation. Five variables are sampled: inductance L , capacitance C , switching frequency f_{sw} , load current I_{load} , and MOSFET on-resistance $R_{DS(on)}$. The implemented CCM analytical model evaluates:

$$\Delta IL = (V_{in} - V_{out})D / (L f_{sw}) \quad (1)$$

$$I_{rms} = I_{load} \sqrt{1 + (\Delta IL / I_{load})^2 / 12} \quad (2)$$

$$\Delta V_{out} = \Delta IL [ESR + 1/(8 C f_{sw})] \quad (3)$$

with $P_{loss}=I_{rms}^2(R_{DS(on)}D+DCR)+0.5V_{in}I_{load}(tr+tf)f_{sw}$ and $\eta=P_{out}/(P_{out}+P_{loss})$, where $P_{out}=5I_{load}$. The analytical implementation uses $ESR=DCR=10$ m Ω and $tr=tf=20$ ns. Latin-hypercube sampling with seed 42 produced 70,000 points.

TABLE II. ANALYTICAL DESIGN SPACE

Quantity	Range / value
L	1–47 μ H
C	10–470 μ F
f_{sw}	100 kHz–1 MHz
I_{load}	0.1–5 A
$R_{DS(on)}$	5–50 m Ω
ESR / DCR	10 / 10 m Ω
$tr=tf$	20 ns

B. Switching-Level LTspice Data Generation

The automation evaluates the first 300 analytical design coordinates rather than a separately drawn input set. For each coordinate, $R_{load}=5/I_{load}$ and duty remains 5/12 while L , C , f_{sw} , and switch R_{on} are patched into a common LTspice schematic. The reference uses a voltage-controlled switch, simplified diode, 1 m Ω inductor series resistance, no explicit capacitor ESR, 1 ns PWM edges, and a 5 ms transient with measurements over 4–5 ms. The residual thus represents the discrepancy between the implemented model fidelities.

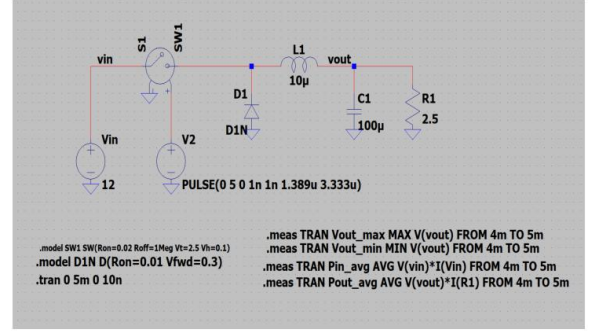


Fig. 2. Automated LTspice reference circuit and original 4–5 ms measurement protocol.

All 300 simulations are generated before filtering. The automation retains rows satisfying $\eta \leq 1$ and $V_{out,max} < 5.5$ V, leaving 285 switching-level rows. These are numerical/operating sanity filters, not an explicit CCM-removal rule.

C. Physics-Guided Features and Residual-Corrected Surrogate

Three physics-derived features augment the five design variables: $Q_{factor}=(5/I_{load})\sqrt{C/L}$, $f_{LC}=1/(2\pi\sqrt{LC})$, and f_{sw}/f_{LC} . The analytical data are split 80:20 with $random_state=42$. Random Forest uses 100 trees; XGBoost uses 200 estimators, depth 6, and learning rate 0.1; and the scikit-learn MLP uses hidden layers (64,32), $max_iter=500$, with standardized inputs and targets.

For each target y , XGBoost supplies the analytical base prediction and the LTspice discrepancy is learned as $r=y_{LTspice}-\hat{y}_{XGB}$. Separate GradientBoostingRegressor models use 50 estimators, $max_depth=2$, $learning_rate=0.05$, $subsample=0.8$, and $random_state=42$. A 75:25 LTspice hold-out split provides a direct test; shuffled five-fold cross-validation measures robustness; and final residual models are refit on all 285 rows before optimization.

$$\hat{y}_{corr}(x) = \hat{y}_{XGB}(x) + \hat{r}_{GBR}(x) \quad (4)$$

TABLE III. CROSS-FIDELITY AND RESIDUAL PERFORMANCE

Target	XGB→LTspice R^2	Corrected hold-out R^2	Stored 5-fold R^2
Efficiency	−2.479	0.301	$0.127 \pm 0.093^*$
Ripple	−0.261	0.728	0.724 ± 0.095
Total loss	0.631	0.956	0.964 ± 0.008

*Efficiency five-fold MAE ≈ 0.38 percentage points; MAE is more informative than R^2 for this narrow-range target.

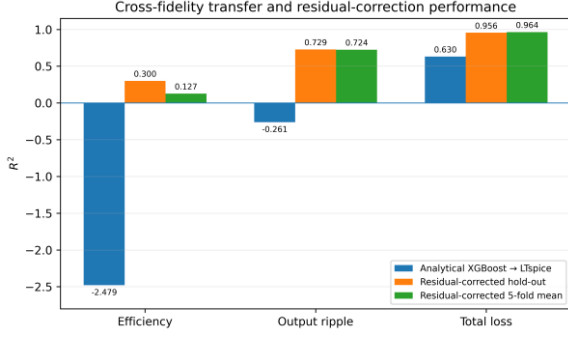


Fig. 3. Analytical-to-LTspice transfer versus residual-corrected performance.

D. NSGA-II Multi-Objective Optimization

1) Extrapolation Diagnostic

NSGA-II [9] minimizes $[-\eta, \Delta V_{out}, P_{loss}]$ at a nominal 2 A load setting, corresponding to $R_{load}=2.5 \Omega$ at the 5 V target. The implementation uses population 100, 100 generations, simulated-binary crossover probability 0.9 with distribution index 15, polynomial mutation with index 20, and seed 42. The initial broad search produced a physically impossible negative ripple prediction and a very large ripple error at another extreme. This failure is retained as evidence that ordinary surrogate test accuracy does not guarantee safe optimizer behavior.

2) Safeguarded Search Within the Empirical LTspice Envelope

The final search uses the cleaned LTspice per-variable extrema, forming an empirical hyper-rectangular envelope rather than a guarantee of dense joint coverage. The code additionally applies $\Delta V_{objective}=\max(\Delta V_{raw}, 1 \mu V)$, so non-negative ripple is partly enforced by an explicit safeguard.

TABLE IV. FINAL NSGA-II SEARCH ENVELOPE

Variable	Final search range
L	2.17–46.6 μH
C	10.1–468 μF
fsw	102.201–999.719 kHz
RDS(on)	5.1–49.9 m Ω

The final run returns 100 surrogate non-dominated candidates. A balanced design is selected after min–max normalizing efficiency, ripple, and loss and minimizing equal-weight Euclidean distance to the ideal vector [maximum efficiency, zero ripple, zero loss]. This selects $L=29.418 \mu H$, $C=158.710 \mu F$, $fsw=103.160$ kHz, and $RDS(on)=5.103$ m Ω .

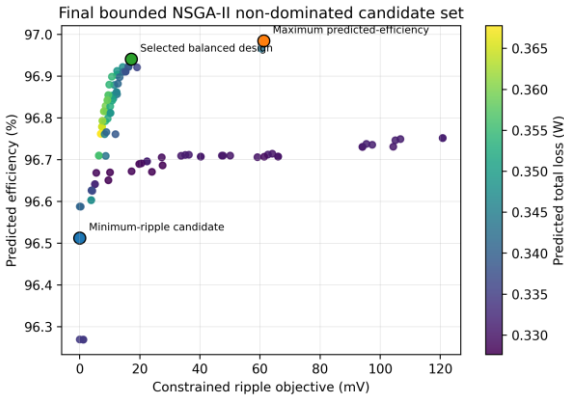


Fig. 4. Final bounded surrogate non-dominated candidate set; ripple objective floored at 1 μV .

IV. RESULTS AND VALIDATION

Three representative candidates are checked using the original automated 4–5 ms protocol: the minimum-ripple extreme, the maximum surrogate-predicted-efficiency extreme, and the normalized-distance balanced design. These points probe different candidate-set regions; they do not imply that all 100 candidates were simulated. Table V lists the candidate parameters and Table VI reports the surrogate-versus-LTspice comparison.

TABLE V. REPRESENTATIVE CANDIDATE DESIGN PARAMETERS

Point	L and C	fsw and RDS(on)
Min-ripple	L = 24.62 μH C = 40.10 μF	fsw = 779.733 kHz RDS(on) = 5.2 m Ω
Max pred.- η	L = 29.47 μH C = 20.75 μF	fsw = 102.224 kHz RDS(on) = 5.1 m Ω
Selected	L = 29.42 μH C = 158.7 μF	fsw = 103.161 kHz RDS(on) = 5.1 m Ω

TABLE VI. SURROGATE PREDICTIONS AND ORIGINAL-PROTOCOL LTSPICE CHECKS

Point	η pred. / LTspice (%)	Ripple obj. / LTspice (mV)
Min-ripple	96.51 / 96.13	0.001† / 0.92
Max pred.- η	96.98 / 96.14	61.21 / 58.78
Selected	96.94 / 96.27	17.08 / 14.65

†0.001 mV is the explicit 1 μV optimization floor. All LTspice values in Table VI use the original 4–5 ms protocol. Efficiency differences are 0.38–0.84 percentage points and ripple differences are 0.92–2.43 mV. The selected balanced candidate measures 96.27% efficiency versus 96.14% for the maximum surrogate-predicted-efficiency candidate; so the latter label refers to surrogate ranking, not measured LTspice ranking.

A. Source-Level Reproduction and Averaging-Window Sensitivity

The saved validation netlists reproduce the candidate variables to formatting precision. Rebuilding the maximum predicted-efficiency candidate from rounded schematic values and measuring over 4–5 ms gives $\eta=96.1438\%$ and $\Delta V_{out}=58.769$ mV, essentially reproducing the stored 96.14% and 58.78 mV values. This provides a direct recoverability check of the original validation procedure.

Averaging-window sensitivity was checked on the same rounded maximum surrogate-predicted-efficiency circuit. The original 4–5 ms window gives $\eta=96.1438\%$ and $\Delta V_{out}=58.769$ mV, whereas 9–10 ms gives $\eta=95.9074\%$ and $\Delta V_{out}=58.749$ mV. Thus the ripple is essentially unchanged while the computed efficiency shifts by 0.236 percentage points. Under the same 9–10 ms manual window, the selected balanced design gives $\eta=96.3017\%$, again exceeding the 95.9074% value of the maximum surrogate-predicted-efficiency design. Table VI is therefore retained strictly as the original fixed 4–5 ms validation protocol; the alternate-window result is reported only as a sensitivity check. Future high-precision efficiency characterization should average over an integer number of switching cycles after confirmed settling.

V. DISCUSSION AND LIMITATIONS

The central result is that analytical-domain accuracy is insufficient evidence for switching-level fidelity. XGBoost reaches approximately $R^2=0.99$ on analytical efficiency and 0.90 on analytical ripple, yet transfers poorly to LTspice. Residual learning recovers useful higher-fidelity structure from only 285 simulations, while optimization reveals a failure mode

that ordinary test metrics can miss. The contribution is hence the audited multi-fidelity workflow and its reliability analysis, not a new ML algorithm.

The study remains simulation-level. The LTspice circuit uses simplified switching elements and is not a transistor-level or physical prototype. Analytical and LTspice parasitic/transition assumptions differ, the empirical per-variable min–max envelope does not ensure dense joint coverage, and the 1 μ V floor explicitly prevents negative ripple objectives. Efficiency and total loss are learned by separate residual models rather than hard-constrained to exact power balance; across the stored Pareto set, their implied efficiency consistency differs by about 0.18 percentage points on average (maximum about 0.49 percentage points). Only three final candidates are checked under the original protocol, the optimization uses a nominal 2 A load setting, and the original requirements file does not pin package versions. A final public repository should also include a version-locked environment in addition to the data, notebooks, netlists, and logs.

VI. CONCLUSION

A source-audited physics-guided multi-fidelity workflow has been demonstrated for buck-converter design. Analytical equations provide inexpensive coverage, and matched LTspice runs quantify the fidelity gap. Target-wise residual learning then corrects the analytical surrogate, while NSGA-II generates bounded trade-off candidates and highlights extrapolation risk. The supplied notebooks, CSV files, netlists, and LTspice logs reproduce the central numerical claims, including the corrected 10.1 μ F capacitance lower bound, explicit 1 μ V ripple floor, watt-valued loss predictions, exact data filters, and representative validation results. Hardware testing, denser high-fidelity sampling, uncertainty-aware search constraints, and physics-consistent multi-output learning are the highest-value extensions.

AUTHOR CONTRIBUTIONS, FUNDING, DATA AVAILABILITY, AND ACKNOWLEDGMENTS

Author contributions: Raushan Kumar conceived the study, developed the analytical and simulation workflow, performed the machine-learning and optimization experiments, analyzed the results, and prepared the manuscript. Funding: No external funding was received for this study. Data availability: The notebooks, processed data, LTspice netlists, and validation logs supporting the reported results are available in a public code repository [12]. Acknowledgments: The author acknowledges the academic environment provided by LNJPIIT; the study was conducted independently. Conflicts of interest: The author declares no conflict of interest.

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